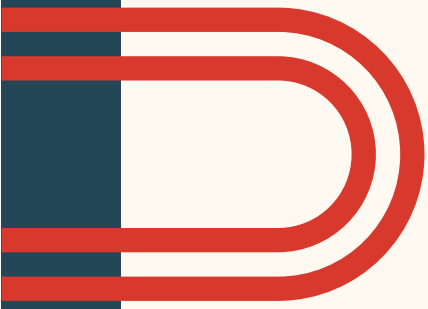




RECIPE AI

SMART INGREDIENT BASED RECIPE FINDER

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INTRODUCTION

PROBLEM DEFINITION

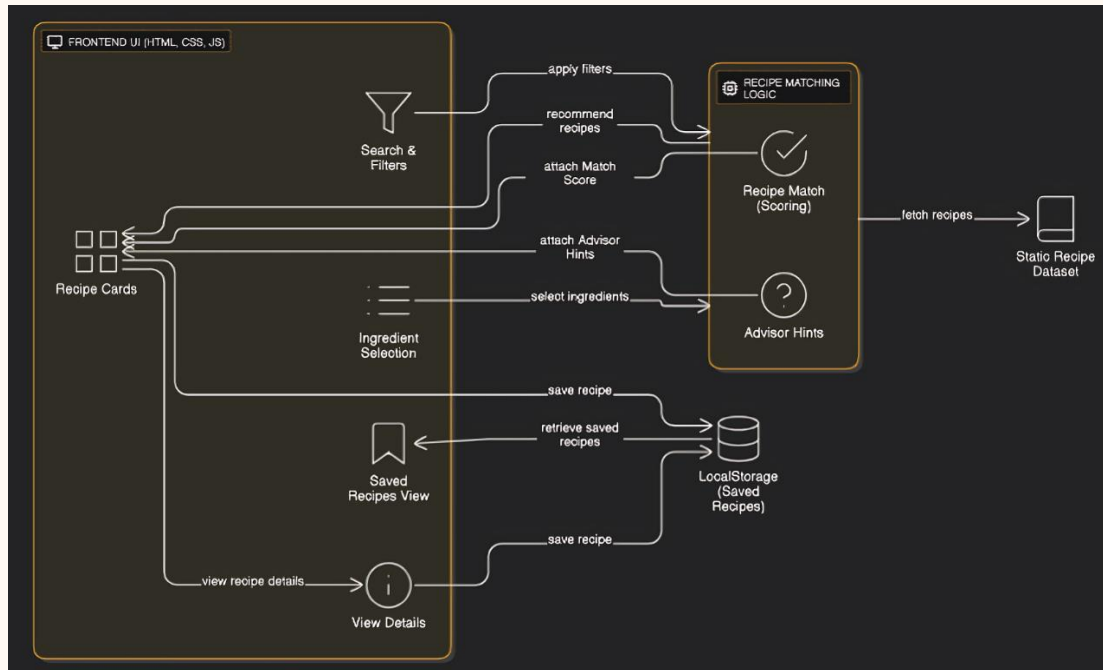
Cooking has become an essential daily activity, yet many people struggle with deciding *what to cook* based on the ingredients they already have at home. Existing recipe resources such as cookbooks or generic websites often require users to search manually, which can be time-consuming and inefficient.

The problem is the **lack of a simple and intelligent system that allows users to quickly discover recipes based on available ingredients, dietary preferences, and time constraints**. Without such a system, users either waste time browsing irrelevant recipes or end up not utilizing the ingredients they already have, leading to food wastage.

OBJECTIVES OF THE PROJECT

- 1.To suggest accurate recipes instantly based on ingredients provided by the user.
- 2.To minimize food wastage by maximizing the use of available household items.
- 3.To provide recipe recommendations tailored to dietary preferences and nutritional needs.
- 4.To design a system that is both simple for users and scalable for larger recipe datasets.

SYSTEM DESIGN



The system is designed with three main layers: Frontend UI, Recipe Matching Logic, and Data Storage.

- The Frontend UI (HTML, CSS, JS) allows users to search, filter, and view recipe cards. Users can also select ingredients, view detailed instructions, and save recipes.
- The Recipe Matching Logic processes filters and ingredient choices. It assigns a match score to rank recipes and provides advisor hints for substitutions or improvements. This logic fetches data from the static recipe dataset.
- The Data Storage layer uses local storage to save user-selected recipes for later retrieval, ensuring quick access without a backend server.

This design ensures personalized recommendations, lightweight storage, and a user-friendly browsing experience.

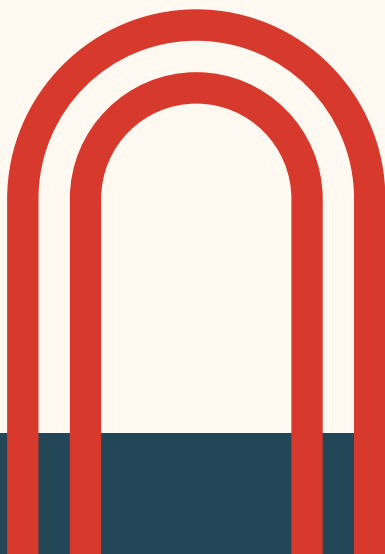
METHODOLOGY

TOOLS AND TECHNOLOGIES

- Front End – HTML , CSS , JavaScript
- Data handling and Storage – Local storage , JSON

STEP BY STEP APPROACH

1. **User Input:** Collect available ingredients, optional filters (cuisine, diet, cooking time).
2. **Normalize Data:** Convert ingredients and filter values to lowercase for consistency.
3. **Filter Recipes:** Apply cuisine, diet, and time filters to the recipe dataset.
4. **Match Ingredients:** Compare user ingredients with recipe ingredients using sets.
5. **Score Recipes:** Calculate match percentage and generate reasoning/hints.
6. **Sort & Select Top Recipes:** Pick the top 3 recipes based on match score.
7. **Display Results:** Render recipe cards with save/view functionality and reasoning.
8. **User Actions:** Save recipes to localStorage, view full recipe modal.
9. **Dynamic Updates:** Update saved recipes sidebar and advisor hints



EXISTING SYSTEMS

- **Cookbooks & printed guides** – physical recipe books organized by cuisine, diet, or occasion.
- **Youtube & video tutorials** – people watch cooking videos and pick recipes manually.
- **Popular recipe apps (Allrecipes, Yummly, Tasty)** - suggest recipes based on keyword search or pre-set filters.

THEIR LIMITATIONS:

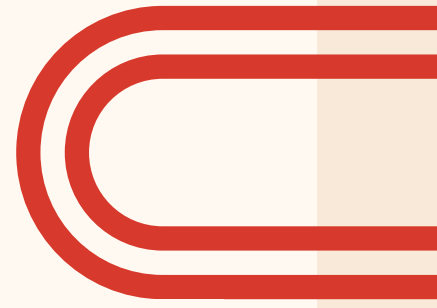
- **Limited ingredient reasoning** – most apps just match keywords, not combinations.
- **No personalized AI suggestions** – minimal adaptation to user preferences.
- **Overwhelming choices** – many recipes, but no smart ranking or match score.
- **Lack of reasoning/explanation** – users don't know *why* a recipe is recommended.

IMPROVEMENT AREAS:

- **Dynamic AI scoring** – rank recipes based on **ingredient match, cooking time, and diet preferences**, not just keyword search.
- **Reasoning transparency** – show **why a recipe fits**, helping users make smarter choices.

Creative substitutions – suggest viable ingredient alternatives to maximize pantry utilization

RESULTS



BENEFITS

- **Reduces Food Waste:** Suggests recipes based on what users already have, minimizing unused ingredients.
- **Saves Time & Effort:** Quickly narrows down recipe options instead of browsing multiple cookbooks or websites.
- **Promotes Culinary Exploration:** Introduces users to new cuisines, cooking techniques, and flavor combinations.
- **Supports Sustainable Living:** By using locally available or leftover ingredients, it encourages smarter consumption habits.

POSSIBLE IMPROVEMENTS

- **Adaptive Personalization:** Incorporate user history and preferences to suggest recipes based on previously selected or saved recipes.
- **Advanced Ingredient Matching:** Support partial matches, ingredient synonyms, and common substitutes (e.g., “bell pepper” = “capsicum”) to increase recipe relevance.
- **Enhanced Scoring Algorithm:** Include additional factors like seasonal availability, nutritional content, and user-defined priorities (e.g., low-calorie, high-protein).

CONCLUSION

- Developed an **AI-powered recipe recommendation system** that matches user-provided ingredients with available recipes.
- **Dynamic scoring and reasoning:** Recipes scored based on ingredient match %, filters (cuisine, diet, time), and AI-generated reasoning.
- **Personalized hints:** AI suggests best recipes based on user's input to reduce waste and optimize cooking choices.

FUTURE VISION

- **Full AI-Powered Kitchen Assistant:** Beyond recommending recipes, the system can suggest weekly meal plans, generate shopping lists, and optimize ingredient usage to minimize wastage.
- **Gamified Cooking Experience:** Reward users for trying new recipes, matching ingredients efficiently, or reducing waste.
- **Ingredient Inventory Tracking:** Keep a virtual pantry that updates as recipes are cooked to suggest what's left or needs replenishing.
- **AI-Powered Cooking Assistant:** Provide step-by-step voice guidance, timers, and tips for a hands-free cooking experience.